

Configuração dos Projetos

Projetos do STM32CUBEmx

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STM32L4 Series

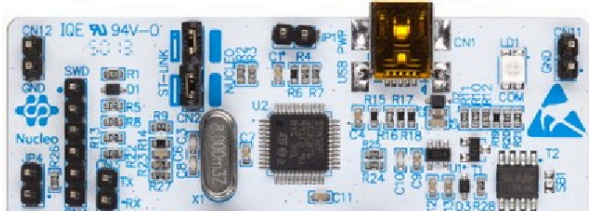
NUCLEO-L476RG

STM32 Nucleo-64 development board with STM32L476RG MCU, supports Arduino and ST morpho connectivity

ACTIVE
Product is in mass production

Part Number : NUCLEO-L476RG
Commercial Part Number : NUCLEO-L476RG

Unit Price (US\$) : 14.0
Mounted Device : STM32L476RGT3



The STM32 Nucleo-64 board provides an affordable and flexible way for users to try out new concepts and build prototypes by choosing from the various combinations of performance and power consumption features provided by the STM32 microcontroller. For the compatible boards, the internal or external SMPS significantly reduces power consumption in Run mode.

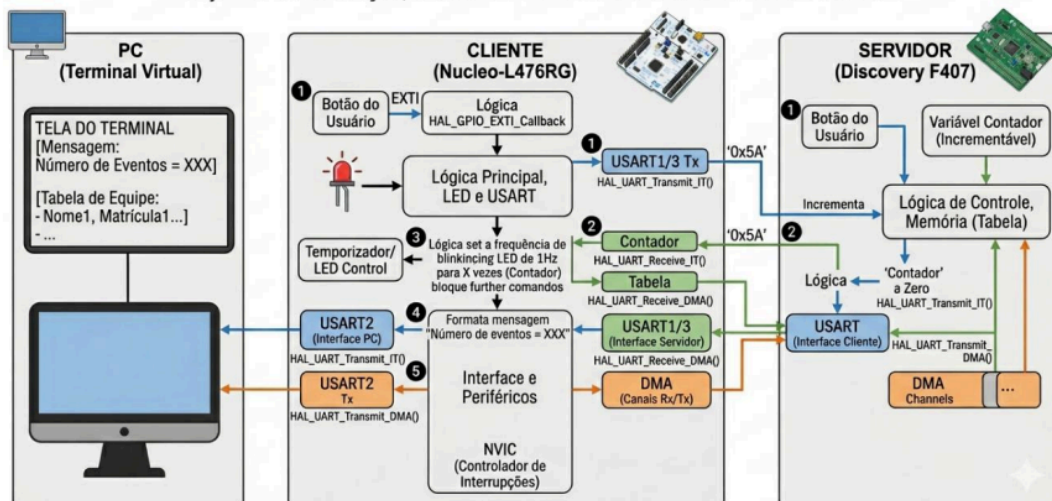
The ARDUINO® Uno V3 connectivity support and

Configuração NUCLEO (L476RG)

O Projeto do professor solicita:

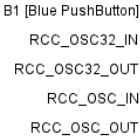
DIAGRAMA DE BLOCOS SIMPLIFICADO: SISTEMA CLIENTE-SERVIDOR E INTERFACE PC

Objetivo: Comunicação, Controle de LED e Transmissão de Tabela via DMA



(Imagem gerada por IA)

 1 model view = **system view**

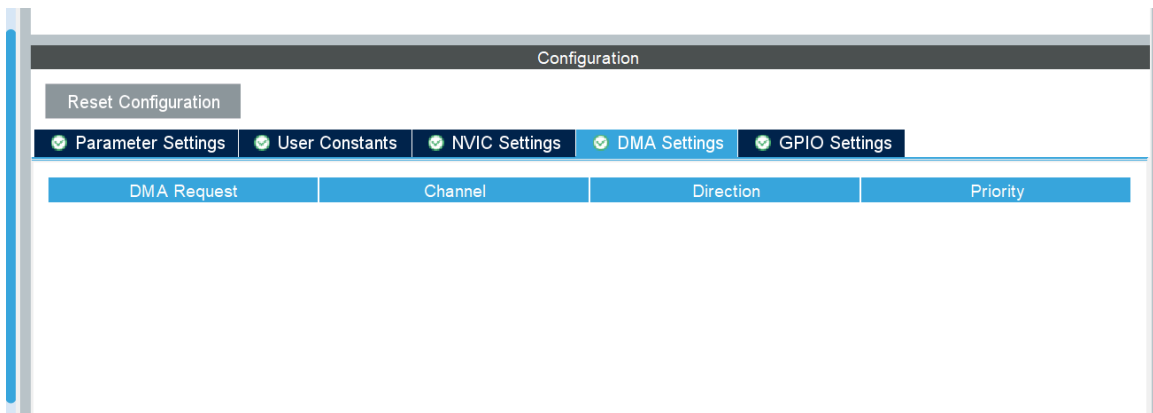
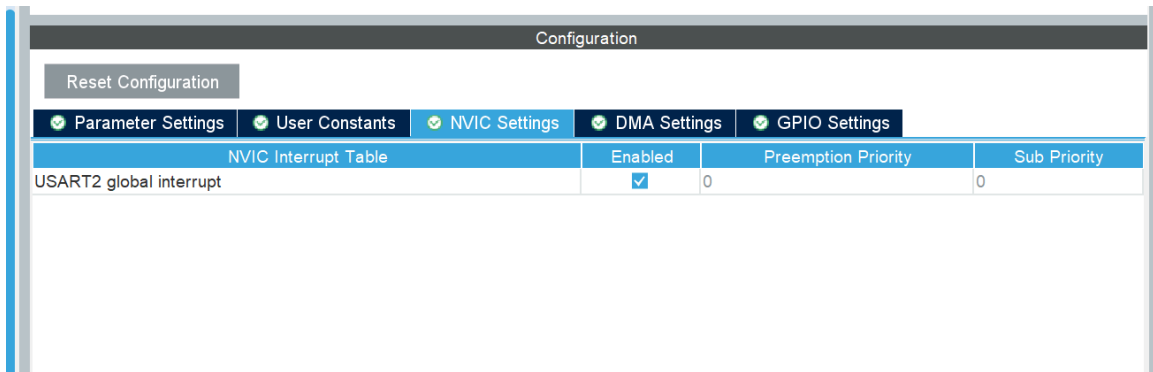
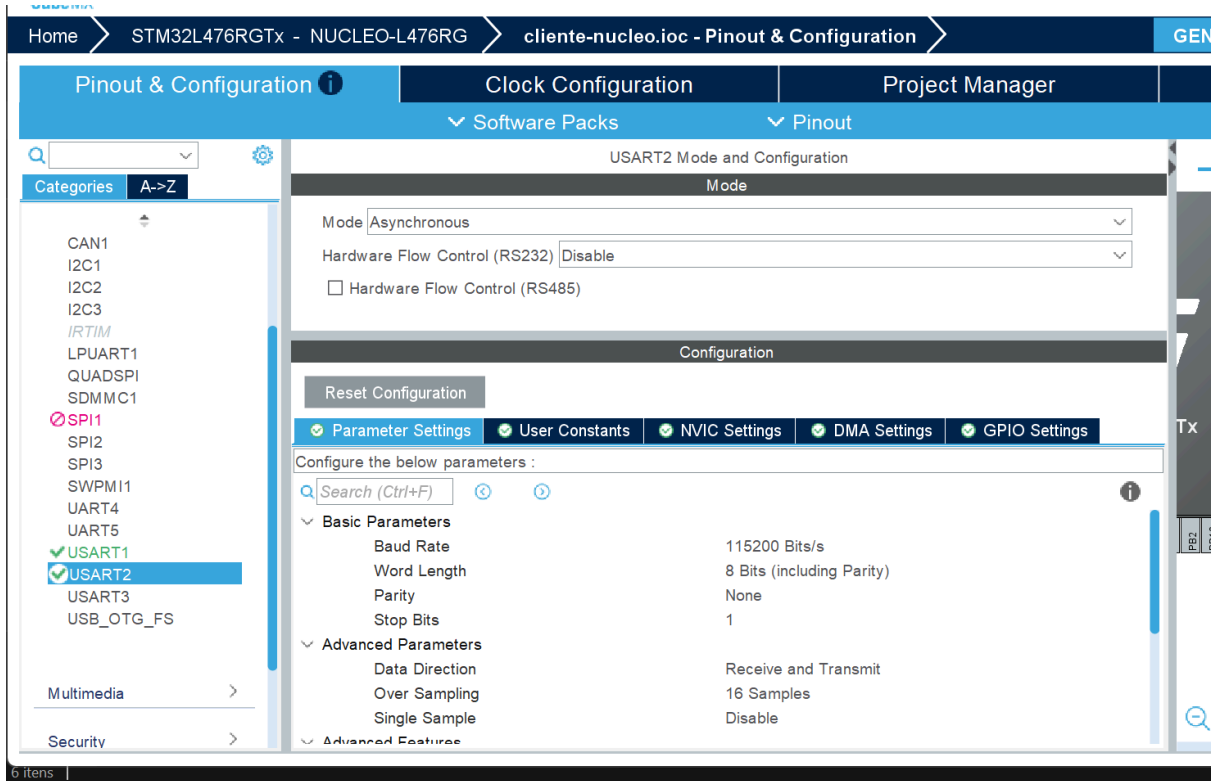


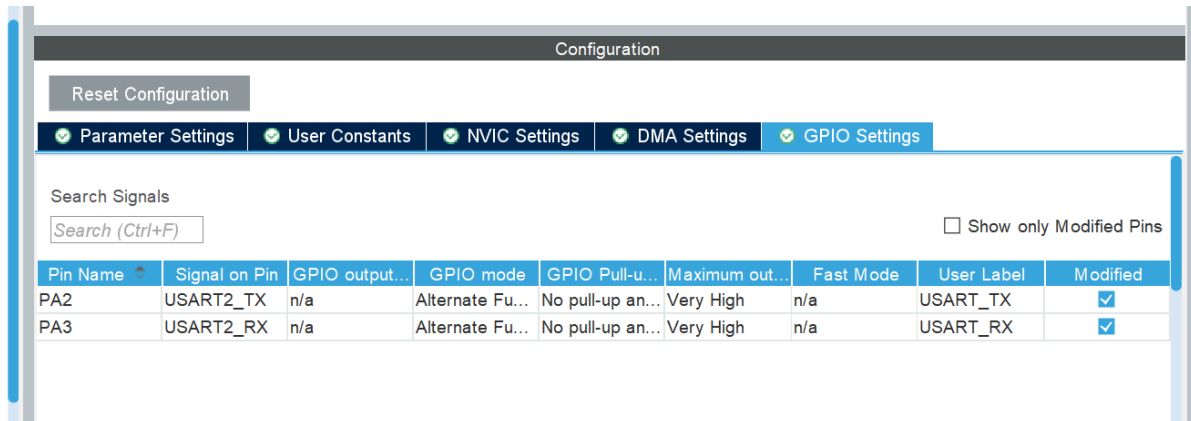
VBAT	
PC13	
PC14-..	
PC15-..	

The screenshot shows the 'NVIC Mode and Configuration' window. The left sidebar has 'NVIC' selected under 'System Core'. The main window has 'Configuration' selected under 'Mode'. A table lists various interrupts with checkboxes for enabling them and input fields for priority. The 'EXTI line[15:10] interrupts' row is highlighted, and its 'Enabled' checkbox is checked. The 'Preemption Priority' is set to 0.

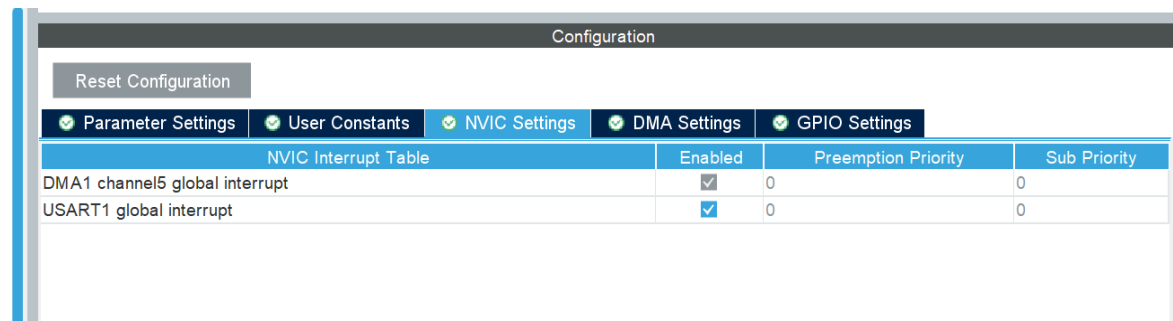
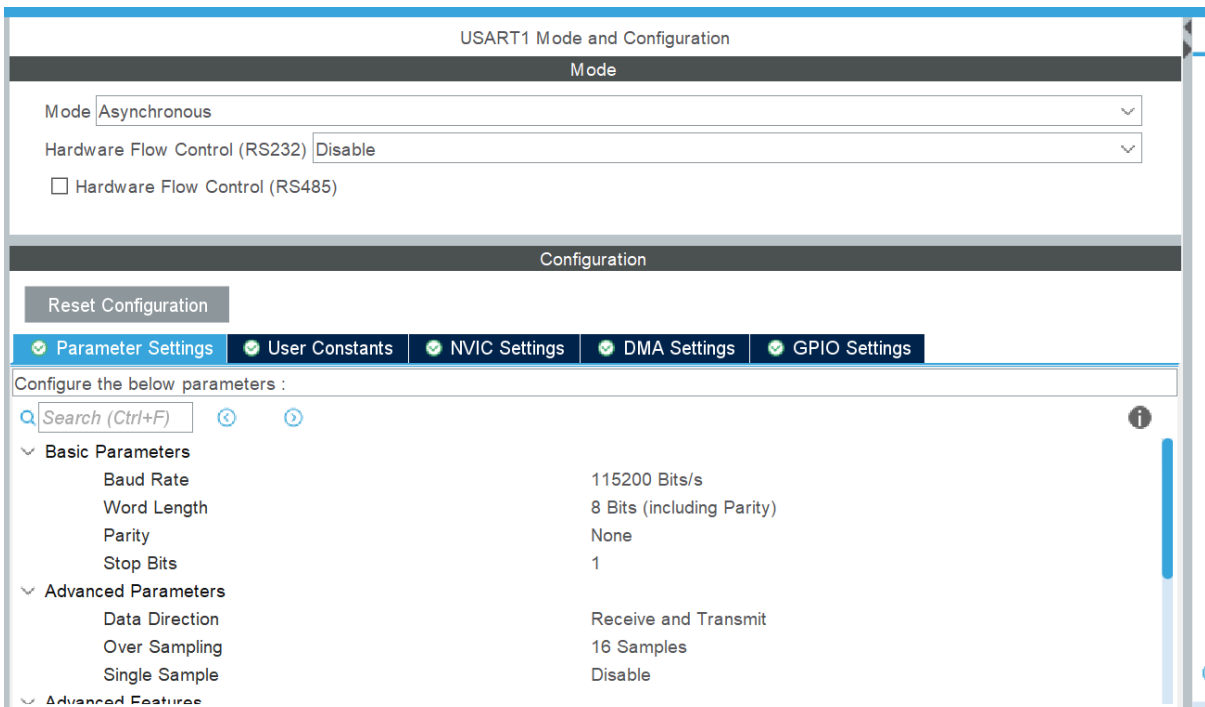
Interrupt	Enabled	Preemption Priority	Sub Priority
Debug monitor	☒	0	0
Pendable request for system service	☒	0	0
Time base: System tick timer	☒	0	0
PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/35/36/37/38	☐	0	0
Flash global interrupt	☐	0	0
RCC global interrupt	☐	0	0
DMA1 channel5 global interrupt	☒	0	0
DMA1 channel7 global interrupt	☒	0	0
USART1 global interrupt	☒	0	0
USART2 global interrupt	☒	0	0
EXTI line[15:10] interrupts	☒	0	0
FPU global interrupt	☐	0	0

Para USART2, que será ligada com a maquina:





Para USART1, que será ligado na Discovery:



Configuration

Reset Configuration

✔ Parameter Settings

✔ User Constants

✔ NVIC Settings

✔ DMA Settings

✔ GPIO Settings

DMA Request	Channel	Direction	Priority
USART1_RX	DMA1 Channel 5	Peripheral To Memory	Low

Add

Delete

DMA Request Settings

Mode

Normal

▼

Increment Address

☐

Data Width

Byte

▼

Peripheral

☐

Memory

☒

Byte

▼

Configuration

Reset Configuration

✔ Parameter Settings

✔ User Constants

✔ NVIC Settings

✔ DMA Settings

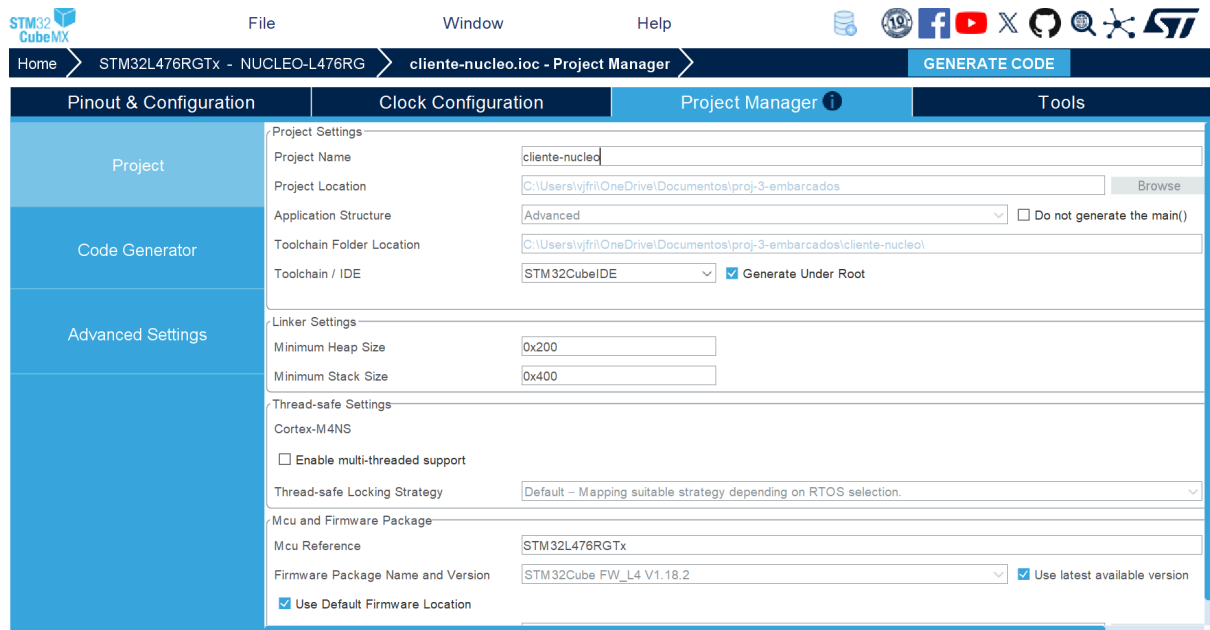
✔ GPIO Settings

Search Signals

Search (Ctrl+F)

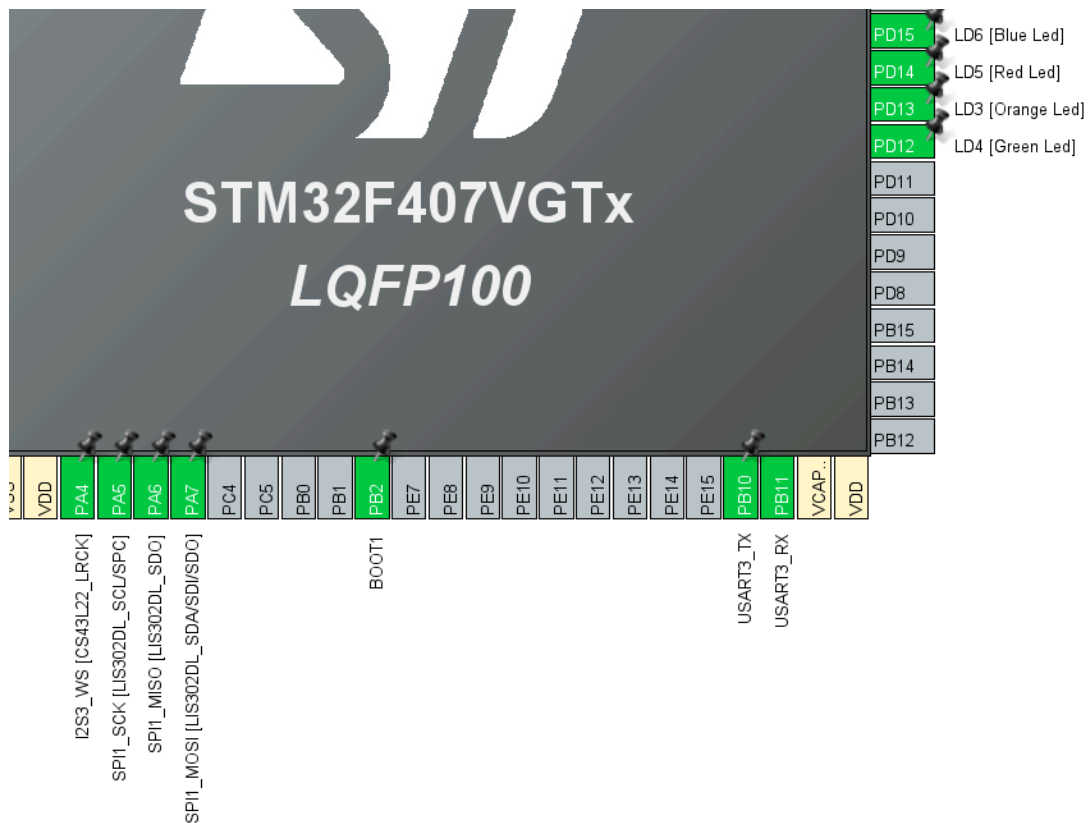
☐ Show only Modified Pins

Pin Name	Signal on Pin	GPIO output...	GPIO mode	GPIO Pull-u...	Maximum out...	Fast Mode	User Label	Modified
PA9	USART1_TX	n/a	Alternate Fu...	No pull-up an...	Very High	n/a		☐
PA10	USART1_RX	n/a	Alternate Fu...	No pull-up an...	Very High	n/a		☐

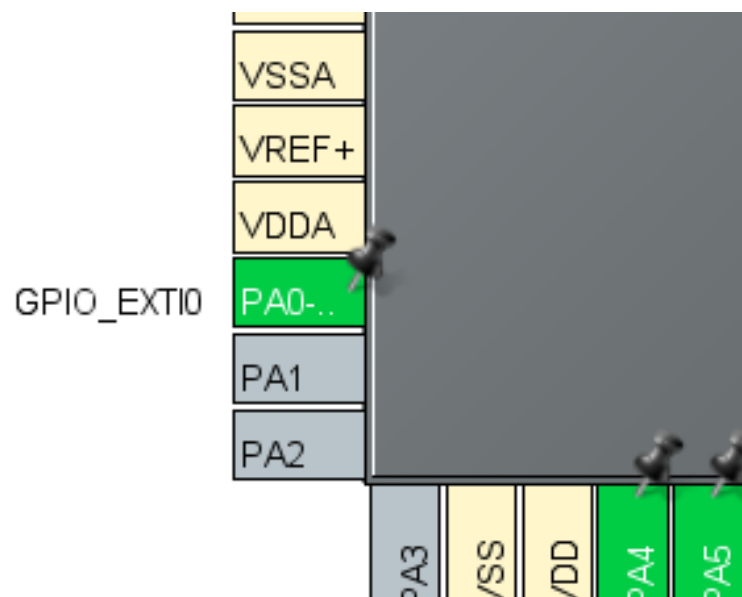
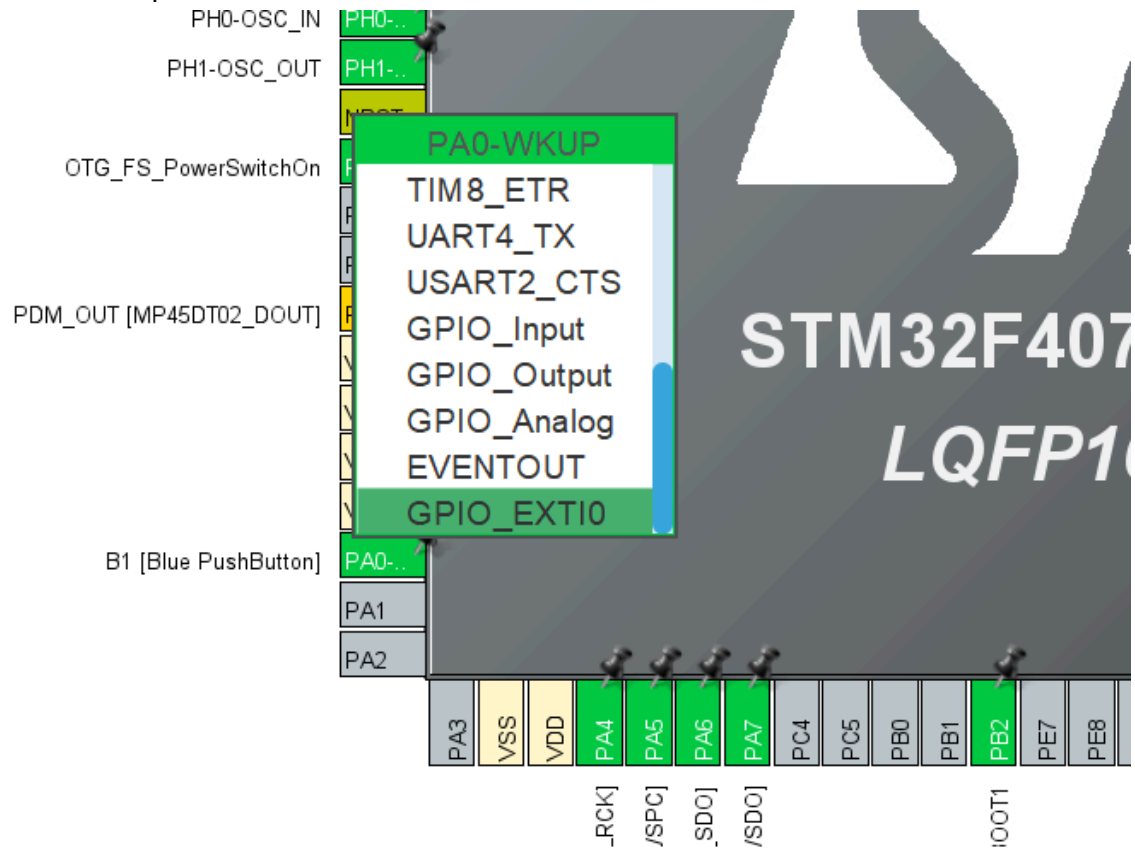


(OBS: sempre gerar código e colocar a IDE correta!)

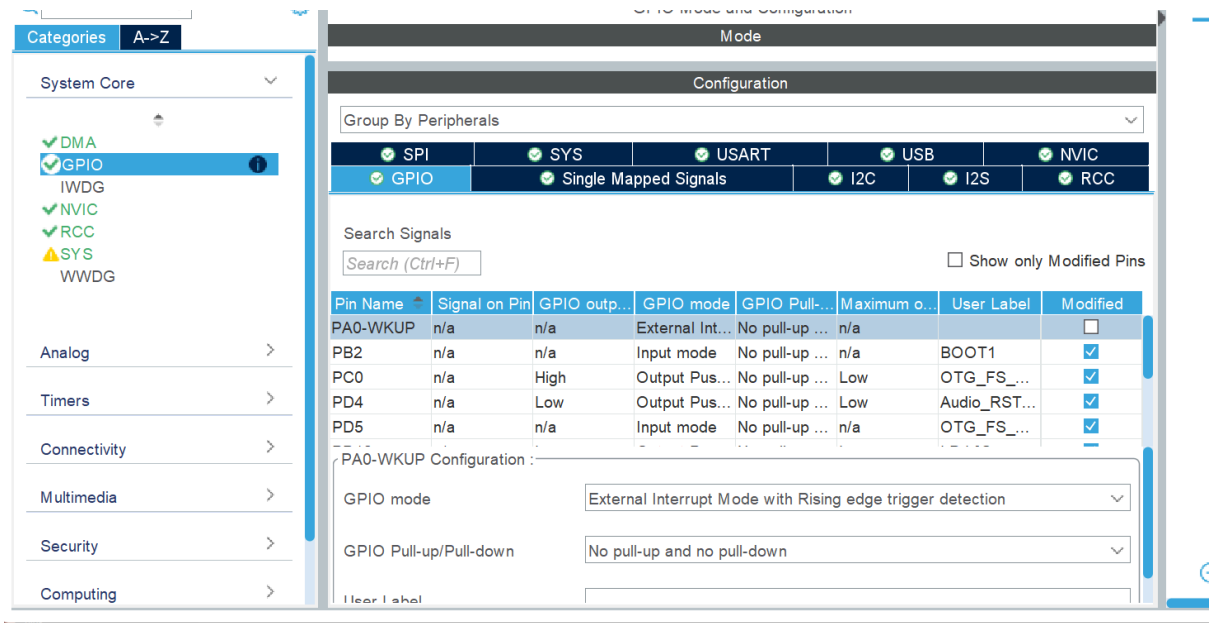
Configuração DISCOVERY (F407)



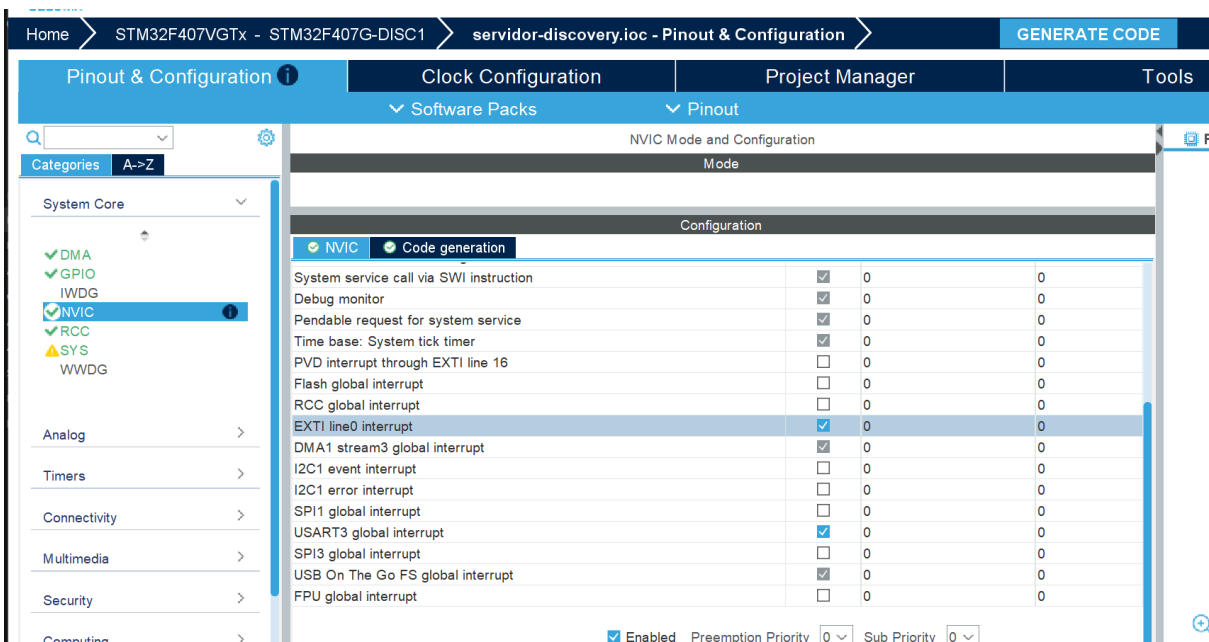
Coloquei o botão para ser o EXTI



conferimos como o GPIO está:



e marcamos a opção do EXTI



COMUNICAÇÃO:

Pinout & Configuration

Categories

Timers

Connectivity

CAN1

CAN2

ETH

FSMC

I2C1

I2C2

I2C3

SDIO

SPI1

SPI2

SPI3

UART4

UART5

UART1

UART2

UART3

UART6

USB_OTG_FS

USB_OTG_HS

Software Packs

Pinout

USART3 Mode and Configuration

Mode

Mode Asynchronous

Hardware Flow Control (RS232) Disable

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

GPIO Settings

Configure the below parameters :

Search (Ctrl+F)

Basic Parameters

Baud Rate 115200 Bits/s

Word Length 8 Bits (including Parity)

Parity None

Stop Bits 1

Advanced Parameters

Data Direction Receive and Transmit

Over Sampling 16 Samples

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

GPIO Settings

NVIC Interrupt Table

Enabled

Preemption Priority

Sub Priority

DMA1 stream3 global interrupt

USART3 global interrupt

Configuration

Reset Configuration

Parameter Settings

User Constants

NVIC Settings

DMA Settings

GPIO Settings

DMA Request

Stream

Direction

Priority

USART3_TX

DMA1 Stream 3

Memory To Peripheral

Low

Add

Delete

DMA Request Settings

Mode Normal

Increment Address

Peripheral

Memory

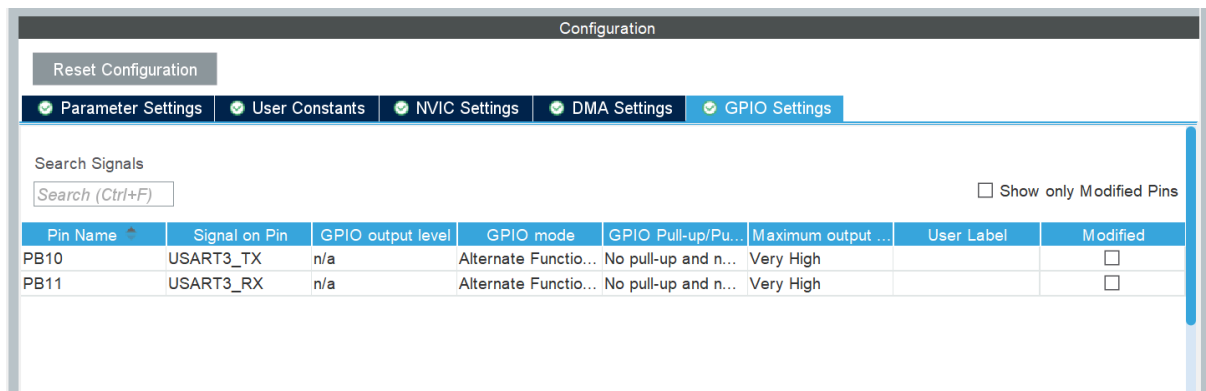
Use Fifo

Threshold

Data Width Byte

Byte

Burst Size



Ligações físicas

Dados:

Núcleo GPIO D2(PA10) [USART1_RX] — jumper roxo----> [USART3_TX] PB10 Discovery.

Núcleo GPIO D8(PA9) [USART1_TX] — jumper cinza----> [USART3_RX] PB11 Discovery.

Alimentação:

Para gravar: ambas ligadas no USB

Para rodar o Projeto:

- 3v3 da nucleo —jumper marrom—> 3v da Discovery
- GND da Nucleo —jumper preto—> GND da discovery

(seguimos nesse modelo como solicitação o prof. em sala.).

Codigo

Codigo Nucleo

```
352 /* USER CODE BEGIN 4 */
353 void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
354 {
355     if(GPIO_Pin == GPIO_PIN_13)
356     {
357         sprintf(log_buffer, "[BTN] Botao pressionado - enviando 0x5A...\r\n");
358         log_msg(log_buffer);
359
360         if(estado == IDLE)
361         {
362             HAL_UART_Transmit_IT(&huart1, &cmd, 1);
363
364             sprintf(log_buffer, "[UART1 TX] Comando enviado.\r\n");
365             log_msg(log_buffer);
366
367             estado = WAITING_SERVER;
368         }
369     }
370 }
```

EXTi do botão

```
400     sprintf(log_buffer, "[TABELA RECEBIDA]...\r\n");
401     log_msg(log_buffer);
402 }
403 /*
404 // reativa recepção do contador
405 HAL_UART_Receive_IT(&huart1, &contador_rx, 1);
406 */
407 }
408
409 void HAL_UART_RxCpltCallback_DMA(UART_HandleTypeDef *huart)
410 {
411     if(huart->Instance == USART1)
412     {
413         sprintf(log_buffer, "[DMA RX] Tabela recebida!\r\n");
414         log_msg(log_buffer);
415
416         sprintf(log_buffer, "[TABELA RECEBIDA]:\r\n%s\r\n", tabela_rx);
417     }
418 }
```

reciver do contador na nucleo e Callback do DMA

```
/* USER CODE BEGIN 3 */
if(estado == PROCESSING)
{
    sprintf(log_buffer, "[STATE] -> PROCESSING\r\n");
    log_msg(log_buffer);

    for(int i = 0; i < contador_rx; i++)
    {
        HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5);
        HAL_Delay(500);
    }

    // garante que termina apagado
    HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);

    sprintf(log_buffer, "[LED] Piscar concluido.\r\n");
    log_msg(log_buffer);
}
```

blink do led, feito por toggle (daí as piscadas pela metade. é um acendendo e outro apagando)

Codigo Discovery

```
433
434 /* USER CODE BEGIN 4 */
435 void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
436 {
437     if(GPIO_Pin == GPIO_PIN_0)
438     {
439         contador++;
440
441         // só toggle, sem delay
442         HAL_GPIO_TogglePin(GPIOD, GPIO_PIN_12);
443     }
444 }
445
446 void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
447 {
448     if(huart->Instance == USART3)
449     {
450         if(rx == 0x5A)
451         {
```

codigo e toggle do led.

```
443     }
444 }
445
446 void HAL_UART_RxCpltCallback(UART_HandleTypeDef *huart)
447 {
448     if(huart->Instance == USART3)
449     {
450         if(rx == 0x5A)
451         {
452             //HAL_UART_Transmit_IT(&huart3, &contador, 1);
453             //HAL_UART_Transmit_DMA(&huart3, (uint8_t*)tabela, sizeof(tabela));
454             HAL_UART_Transmit(&huart3, &contador, 1, 100);
455             HAL_Delay(10); // pequeno delay
456
457             //HAL_UART_Transmit(&huart3, (uint8_t*)tabela, sizeof(tabela), 100);
458             uint16_t tamanho = strlen(tabela);
459             HAL_UART_Transmit(&huart3, (uint8_t*)&tamanho, 2, 100);
460             HAL_UART_Transmit(&huart3, (uint8_t*)tabela, tamanho, 100);
461         }
462     }
463 }
```

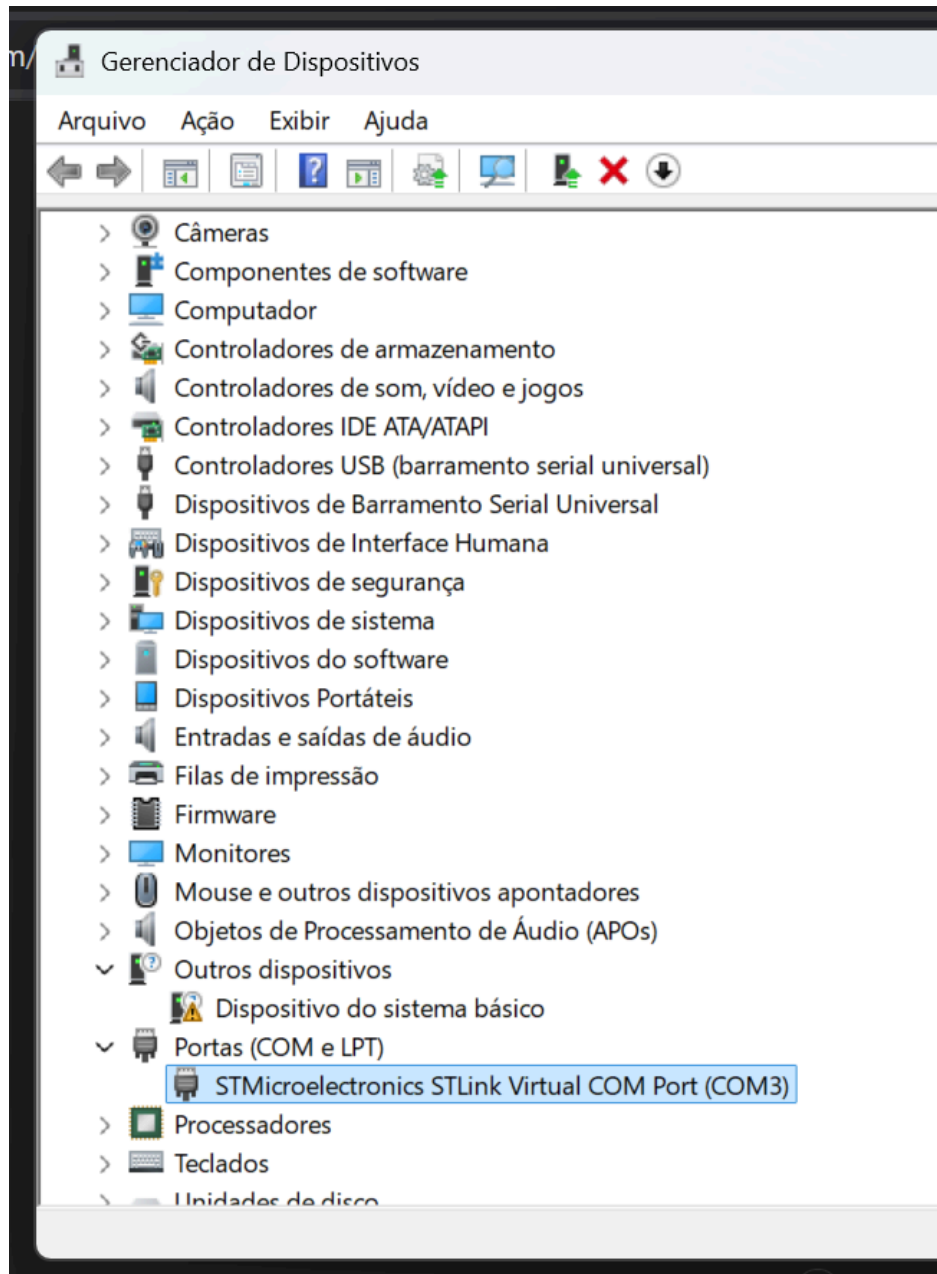
recebe a senha e faz o envio do contador e tabela.

Teste

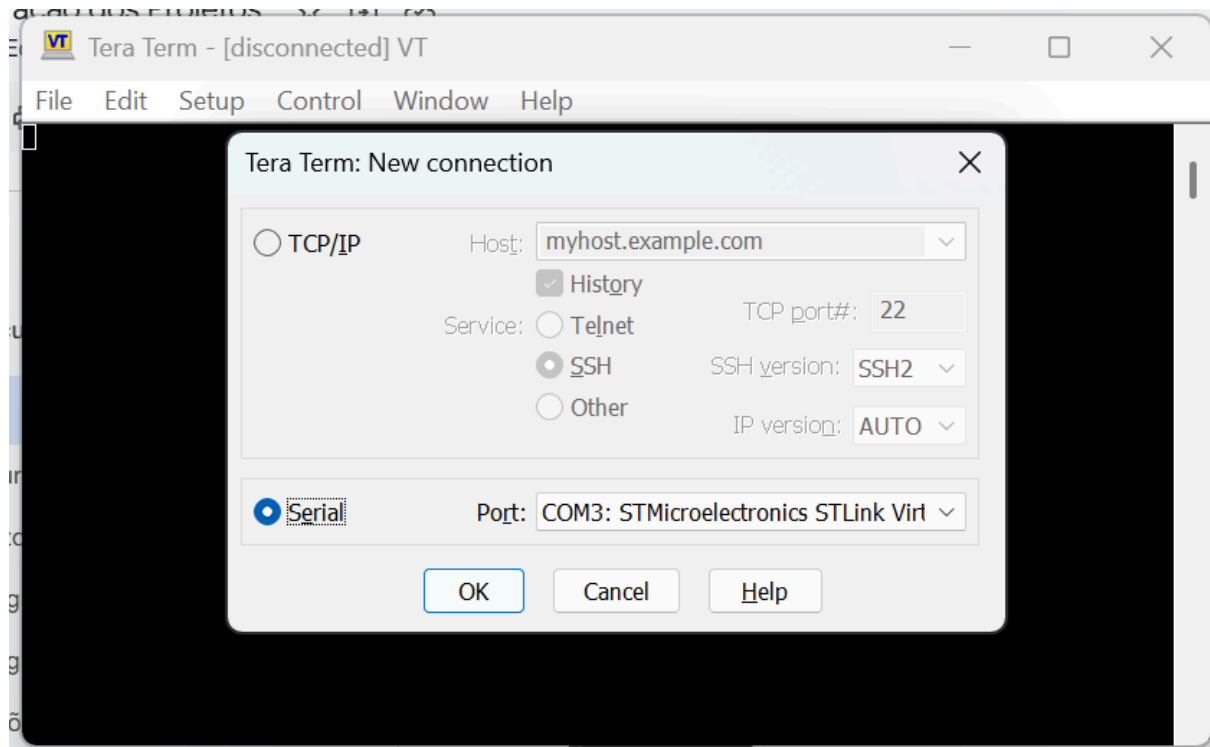
drivers: <https://www.st.com/en/development-tools/stsw-link009.html>

Advanced serial port terminal: <https://www.com-port-monitoring.com/serial-port-terminal/>

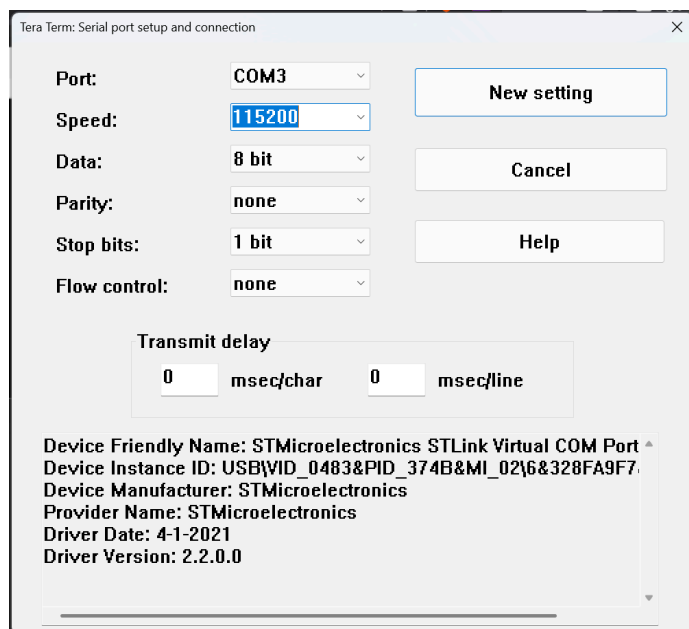
1. Localize a porta COM



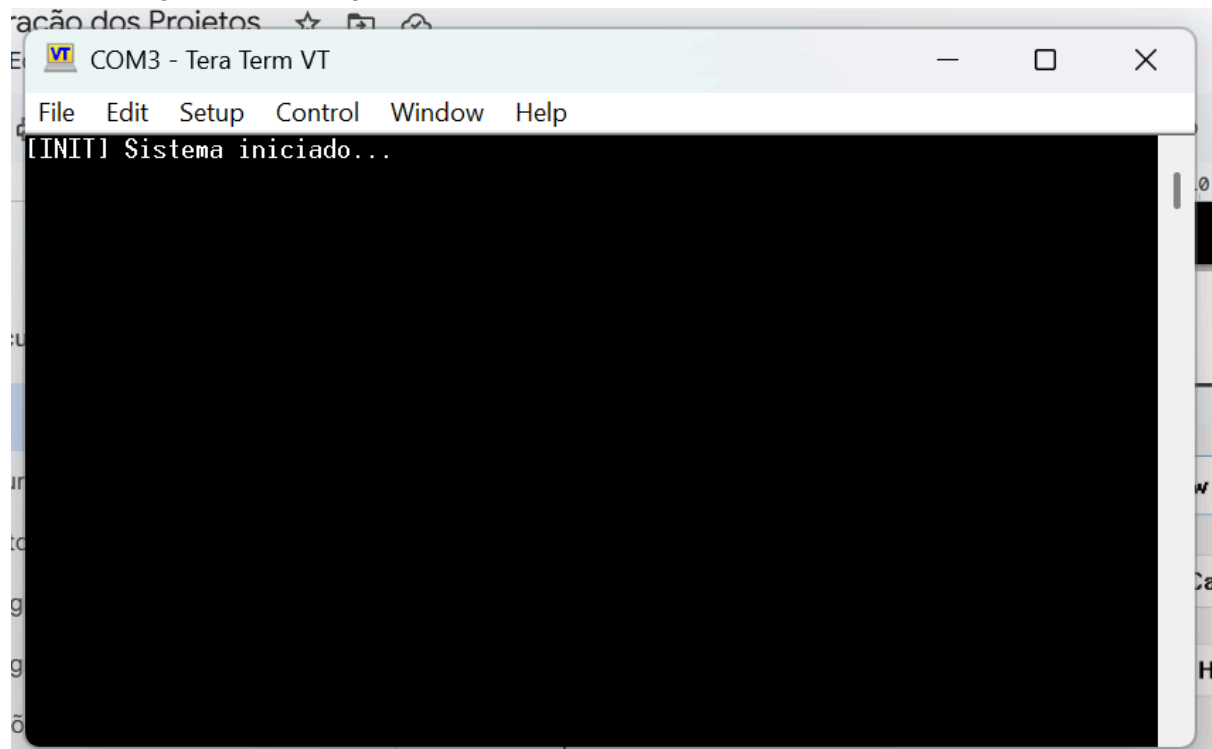
2. Preencha a new session



3. baud rate // speed



4. código de inicialização



5. qnt de botões apertadas recebido

```
[INIT] Sistema iniciado...  
[BTN] Botao pressionado - enviando 0x5A...  
[UART1 TX] Comando enviado.  
[UART1 RX] Contador recebido: 6  
[DMA] Recebendo tabela...  
[STATE] -> PROCESSING  
[LED] Piscar concluido.  
Numero de eventos = 6
```